



A multi-rate hierarchical fault-tolerant adaptive model predictive control framework: Theory and design for quadrotors[☆]

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ABSTRACT

This paper presents the design and stability analysis of a dual-loop hierarchical controller for the trajectory tracking of quadrotors subject to unexpected actuator faults. The hierarchical controller, developed based on a dual-time-scale model decomposition, consists of two loops of control: the outer-loop translation control and the inner-loop rotation control. For translation control, we propose a novel fault-tolerant Lyapunov-based model predictive control strategy with the integration of an adaptive parameter estimator. For rotation control, feedback linearization and adaptive estimation are employed and a fault-tolerant control law is designed. The two loops of the developed hierarchical control system are both implemented in a sample-and-hold scheme with dual sampling rates — the outer-loop is sampled several times slower than the inner-loop. With the inter-sample behavior and the interconnection between the translational and rotational dynamics taken into account, the closed-loop stability of the dual-loop and dual-rate control system is rigorously proven using singular perturbation theory. Sufficient stability conditions are established, based on which control parameters and sampling periods can be tuned jointly such that a trade-off between computation efficiency and control accuracy can be attained. Results of numerical simulations are provided to demonstrate the effectiveness of the proposed control design in trajectory tracking and fault tolerance.

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1. Introduction

The study of unmanned aerial vehicles (UAVs) has grown considerably in the past few decades, motivated by their distinct advantages in executing tasks in dangerous and inaccessible environments (Phang, Li, Yu, Chen, & Lee, 2014). Quadrotors, a typical kind of rotary-wing UAVs that can take off and land vertically, are particularly suited for a wide range of potential applications such as surveillance, transport, rescue, mapping, etc. (Lan, Lai, Lee, & Chen, 2021). For these applications to emerge, the development of autonomous control algorithms for quadrotors is essential, which raises several challenges. First, the developed controller should provide satisfactory closed-loop performance with rapid response and robustness against model uncertainties and external disturbances (Hua, Hamel, Morin, & Samson, 2013). In addition, input constraints, usually resulting from the physical limitations of actuators, should be considered to ensure the

feasibility of the obtained control actions. Moreover, since such UAVs commonly work in complex and hazardous environments, the safety of vehicles and expensive onboard devices may be threatened. It is thus highly desirable that the control system is more reliable with the ability to accommodate potential faults.

When it comes to the control design for quadrotors, it must be taken into consideration that rotary-wing-based aerial vehicles are under-actuated and highly nonlinear mechanical systems. These special features render the control design for quadrotors even more challenging since control techniques developed for fully-actuated and linear systems cannot be directly applied. The stabilization or trajectory tracking problem of quadrotors has been addressed in various ways in the literature, ranging from classical linear control schemes (Bouabdallah, Noth, & Siegwart, 2004; Budiyo & Wibowo, 2007) to more elaborate nonlinear control solutions (Chen, Jiang, Zhang, Jiang, & Tao, 2016; Hauser, Sastry, & Meyer, 1992; Labbadi & Cherkaoui, 2019; Lee, Kim, & Sastry, 2009; Madani & Benallegue, 2006; Razmi & Afshinfar, 2019). Linear controllers are developed based on linear approximation of the dynamics, so their control performance can only be guaranteed within a restricted flight domain. By comparison, nonlinear solutions are developed based on the nonlinear dynamics, yielding controllers with enlarged domains of stability. To deal with the under-actuated dynamics of quadrotors, some

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control strategies are built on dynamic extensions by incorporating the motor's dynamics (Hauser et al., 1992; Lee et al., 2009). The most notable drawback of such schemes is that the involved high-order derivatives increase the sensitivity to external disturbances and sensor noises. Also, the implementation requires the measurement of the motor's states (Hua et al., 2013), which is not available in most cases. The best solution to address the under-actuated issue might be to apply hierarchical control, where cascade controllers are developed successively to control the translational and rotational motions separately. With this control architecture, many nonlinear control methods have been applied to quadrotors such as sliding mode control (Chen et al., 2016; Razmi & Afshinfar, 2019), backstepping control (Labadi & Cherkaoui, 2019), and predictive control (Eskandarpour & Sharf, 2020; Wang, Pan, Shi, Hu, & Zhao, 2021; Zhang, Shi, & Sheng, 2021).

Hierarchical control is developed upon model decomposition by neglecting the interaction between different control loops. Design based on the decomposed model may result in a system far from its desired performance. As for a dual-loop hierarchical quadrotor control system, the overall system stability requires the control gains of the two loops to be chosen appropriately such that the closed-loop rotational dynamics converge faster than the translational dynamics. However, tuning the control parameters is not an easy task. Singular perturbation theory (Khalil, 1992; Kokotović, Khalil, & O'reilly, 1999) provides an effective tool to quantify how "high" the inner-loop rotation control gain should be in view of stability. Through a time-scale separation, it formalizes the control design and the stability analysis in two steps: the first step is to design controllers based on the decomposed model; the second step is to analyze the closed-loop stability by examining the full dynamics with the disregarded interaction taken back into consideration. Results available in the literature that rigorously address the stability issue of hierarchical quadrotor control design are few (Bertrand, Guénard, Hamel, Piet-Lahanier, & Eck, 2011; Esteban, Gordillo, & Aracil, 2013; Pérez-Alcocer, Moreno-Valenzuela, & Miranda-Colorado, 2016). All of them concentrate on continuous-time control without considering the sampled-data nature of flight control systems.

In practical implementation, the quadrotor control system is a typical sampled-data system: the dynamic system evolves in continuous time, whilst the control action is implemented by digital devices that can only process discrete-time signals (Grizzle & Kokotovic, 1988). In most of the existing work on quadrotor control, the sampling period is always assumed to be sufficiently small so that inter-sample behaviors can be ignored. However, we know that small sampling periods require faster and hence more expensive hardware. For many advanced control algorithms with high computational complexities, performing the calculation within such a small time interval may not be practical. In this work, the closed-loop sampled-data quadrotor control system is evaluated rigorously by taking into account the uncontrolled inter-sample system evolution (Zaccarian, Teel, & Nešić, 2003). Explicit stability conditions in terms of sampling periods are established, based on which how the selection of sampling periods would affect the system performance can be quantified. Additionally, the hierarchical control system developed in dual time-scales is also sampled with dual sampling rates — the outer-loop control is updated several times slower than the inner-loop control. Through the dual-rate hierarchical architecture and the theoretical analysis, a larger sampling period that allows for more computationally intensive control methods can be used while the system performance can be still guaranteed.

Further, another problem that arises with the hierarchical control design is that the simplified translational subsystem is

non-affine in control. A common solution in literature is a two-step control design: the first step develops a newly defined virtual control vector including the actual control inputs; the second step calculates the actual inputs, which are the desired rotation angles and the required thrust force, from the control vector obtained in the first step. Model predictive control (MPC) can be a more appropriate alternative for translation control design. Different from other conventional control methods that design analytical feedback control policies offline, MPC determines the control action at each sampling instant by solving a finite horizon optimal control problem online (Li & Shi, 2017; Mayne, 2014). Therefore, applying MPC to the translation control of quadrotors not only brings a certain degree of optimal tracking performance but also determines the actual inputs at one shot. In literature, there are a number of works that apply linear (Eskandarpour & Sharf, 2020; Ma, Xia, Li, & Chang, 2016) or nonlinear MPC (Zhang et al., 2021) to deal with the multirotor control challenges (Shi & Zhang, 2021). For the sake of closed-loop stability, common to all these proposed methods is the need to add a terminal penalty cost and a terminal constraint to the optimization problem of MPC. However, for quadrotors with nonlinear dynamics, the approximated terminal region is usually small, so a longer prediction horizon is required for a sufficiently large region of attraction. A large prediction horizon, though, means a high computational burden, thus preventing the applications of such type of MPC in quadrotors with fast real-time processes. Lyapunov-based MPC is another MPC formulation that was first proposed in Mhaskar, El-Farra, and Christofides (2005). It introduces an additional contractive constraint to delimit a minimal decay rate of the control Lyapunov function. Therefore, stability can be guaranteed without the requirement of terminal constraints, which brings the freedom of choosing a shorter prediction horizon and thereby the advantage of a smaller computation amount. Note that the contractive constraint is only required to be met for the first step of the prediction horizon and its implementation requires few extra calculations, so it will not significantly increase the computational complexity of solving the MPC problem. Lyapunov-based MPC methods have been widely applied in the literature to industry process control (Das & Mhaskar, 2018; MacKinnon, Ramesh, Mhaskar, & Swartz, 2022), and underwater vehicles (Shen & Shi, 2020, 2022; Shen, Shi, & Buckham, 2018; Shi, Shen, Wei, & Zhang, 2023; Wei, Shen, & Shi, 2019). In Wang et al. (2021), Wu, Cai, Zhao, Zhu, and Wang (2017), it is utilized for quadrotor control. However, they only include 2-dimensional translation control for horizontal motion.

To achieve reliable flight, compensating potential faults is significant for quadrotor control, which essentially requires an effective fault-tolerant control (FTC) scheme. MPC itself retains a potential fault-tolerant capability, forming the basis of passive FTC. Passive fault-tolerant MPC either makes use of the inherent robustness of robust MPC (Mhaskar, 2006; Sheikhabaei, Alasty, & Vossoughi, 2018; Yu, Zhang, Minchala, & Qu, 2013) or the inherent self-reconfiguration capability of constrained MPC (Maciejowski, 1998, 1999). The limitation of passive strategies is that the fault effect cannot be entirely compensated due to the ineliminable discrepancy between the real dynamics and the model used for prediction. Active FTC instead reacts to faults by actively recovering the system performance, thus being able to cope with a wider range of faults. The proposed active fault-tolerant MPC schemes are built either on the real-time fault detection and isolation (FDI) (Gopinathan, Boskovic, Mehra, & Rago, 1998; Prodan, Zio, & Stoican, 2015) or online identification that extracts the post-failure parameters from input and output data (Deshpande, Patwardhan, & Narasimhan, 2009; Ferranti, Wan, & Keviczky, 2019; Huang, Naghdy, & Du, 2017; Patan & Korbicz, 2012; Xiao & Liu, 2020). With the knowledge of faults,

reconfiguration to the MPC problem can be made by modifying the prediction model and constraints (Deshpande et al., 2009; Ferranti et al., 2019; Huang et al., 2017; Patan & Korbicz, 2012), or changing the control configuration (Gopinathan et al., 1998; Tao, Zhao, Du, Cheng, & Zhu, 2019; Xiao & Liu, 2020). However, many existing works do not discuss the closed-loop stability even under nominal conditions. The rest ensure stability only under the ideal condition where faults have been correctly isolated or their magnitudes have been accurately estimated. No studies exist yet that addressed the transients and steady-state errors of the fault estimation in the overall system evaluation.

This paper presents the design and stability analysis of a dual-loop hierarchical control system for quadrotors subject to unexpected actuator faults. Translation control and rotation control are successively designed by considering a dual-time-scale decomposition of the translational and rotational dynamics. For the design of the translation controller, we propose a novel fault-tolerant Lyapunov-based model predictive control scheme based on the introduction of an adaptive parameter estimator. For rotation control, we integrate feedback linearization and adaptive parameter estimation. The developed translation control and rotational control are all implemented in a sample-and-hold fashion with different sampling rates. By using singular perturbation theory, the closed-loop stability of the dual-loop and dual-rate quadrotor control system is proven with the sufficient stability conditions provided. Results from numerical simulations are given to illustrate the trajectory tracking and fault-tolerance performance of the proposed design. The main contributions of this paper are four-fold:

- This paper presents a novel adaptive fault-tolerant MPC scheme by combining the Lyapunov-based MPC framework with adaptive estimation. Both multiplicative and additive actuator faults are addressed. The proposed control design can also be applied to a class of nonlinear systems with input constraints and unexpected actuator faults for achieving fault tolerance and optimal control performance.
- The proposed dual-loop and dual-rate control design provides a more practical way to apply the computation-intensive MPC to real-time flight control. With the dual-loop control architecture, MPC is applied to the outer-loop that only involves translation control so that the computation amount for solving the MPC problem is greatly reduced. Under the dual-rate sampling setup, the outer-loop is sampled with a slower sampling rate so that the burden of solving the MPC problem per iteration is mitigated. On the other hand, the inner-loop is sampled with a higher rate, limiting the uncontrolled inter-sampled behaviors and thereby ensuring the overall control performance.
- The proposed MPC-based scheme is able to determine the translation control actions in one step. Other existing solutions, such as the control strategies proposed in Bertrand et al. (2011), Pérez-Alcocer et al. (2016), Voos (2009), require two steps – a virtual control vector is firstly designed and actual control inputs are solved from that. Our approach avoids the singularity problem that might be caused by solving the nonlinear transcendental function in the second step.
- The closed-loop stability is evaluated rigorously. By taking account of the interconnection between subsystems and the inter-sample system evolution, we explicitly characterize the relationship between the closed-loop performance and the selection of control parameters and sampling periods. The obtained sufficient stability conditions specify the maximal singular perturbation parameter, thereby quantifying how “high” at least the control gain of the inner-loop should

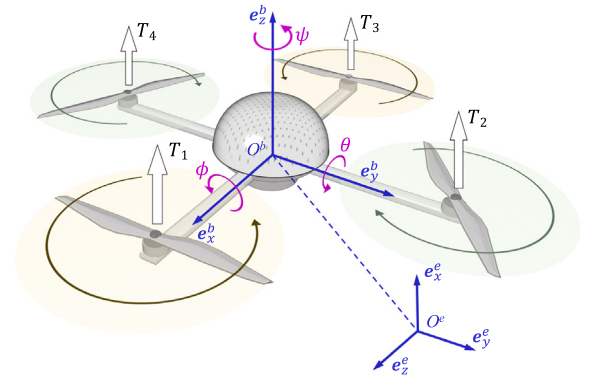


Fig. 1. Schematic and reference frames of a quadrotor.

be for ensuring the overall stability. Following the stability conditions, we can also select the designed parameters together with the sampling periods such that practicable computation effort and guaranteed closed-loop performance can be attained simultaneously.

The rest of this paper is organized as follows. In Section 2, we present the mathematical model of the quadrotor dynamics with constraints and actuator faults and then point out the control objectives. Section 3 decomposes the quadrotor dynamics in two time-scales by using singular perturbation theory. Section 4 presents the main results of this work: a dual-loop and dual-rate hierarchical control system with the outer-loop translation controller and the inner-loop rotation controller developed successively based on the decomposed translational and rotational dynamics model. In Section 5, the closed-loop stability of the full dynamics is rigorously analyzed with the inter-sample behaviors, and the interconnection between the translational and rotational dynamics taken into consideration. In Section 6, simulation results are provided to demonstrate the effectiveness of the proposed control framework and algorithms. Concluding remarks are made in Section 7.

In this paper, notation $\|\cdot\|$ is the standard Euclidean norm. $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ represent the minimal and maximal eigenvalues of a matrix.

2. Problem formulation

2.1. Mathematical modeling

2.1.1. Quadrotor dynamics

The quadrotor we consider is a six-degree-of-freedom rigid-body aerial vehicle moving in three-dimensional space. As shown in Fig. 1, a typical quadrotor is equipped with four rotors symmetrically distributed with an equal distance to the center. The four rotors, set to spin in opposite directions in two pairs, generate upward-lifting forces T_1 , T_2 , T_3 , and T_4 . By manipulating the spinning speed of each rotor, the force and torque required for driving the vehicle to perform certain translation and rotation behaviors can be produced.

To represent the translational and rotational motions, two reference frames are introduced: one is an earth-fixed inertial frame with origin O^e and axes e_x^e , e_y^e , e_z^e ; the other is a body-fixed frame with origin O^b and axes e_x^b , e_y^b , e_z^b . The position of the quadrotor is represented by the coordinate values of the center of mass in the earth-fixed frame, denoted as $\mathbf{p} = [x \ y \ z]^T$. The linear velocity of the vehicle with respect to the body-fixed frame is denoted as $\mathbf{v} = [v_x \ v_y \ v_z]^T$. The attitude of the quadrotor, describing the rotation of the body-fixed frame with respect to the earth-fixed

frame, is denoted by Euler angles as $\mathbf{q} = [\phi \ \theta \ \psi]^\top$, in which ϕ, θ, ψ are the roll, pitch and yaw angles. $\boldsymbol{\omega} = [\omega_x \ \omega_y \ \omega_z]^\top$ is a vector of angular velocities with respect to the body-fixed frame. By using the Lagrangian method, the dynamic model of the quadrotor around a hovering point can be established (Altug, Ostrowski, & Mahony, 2002; Xiong & Zheng, 2014), as represented by the following differential equation sets:

$$\begin{cases} \dot{\mathbf{p}} = \mathbf{v} \\ m\dot{\mathbf{v}} = -m\mathbf{g} + \mathbf{r}(\mathbf{q})T \end{cases} \quad (1a)$$

$$\begin{cases} \dot{\mathbf{q}} = \boldsymbol{\omega} \\ \mathbf{I}\dot{\boldsymbol{\omega}} = -\boldsymbol{\omega} \times \mathbf{I}\boldsymbol{\omega} + \boldsymbol{\tau} \end{cases} \quad (1b)$$

where m is the vehicle mass; $\mathbf{g} = [0 \ 0 \ g]^\top$ with g being the gravitational acceleration;

$$\mathbf{r}(\mathbf{q}) = \begin{bmatrix} \sin \theta \cos \psi \cos \phi + \sin \psi \sin \phi \\ \sin \theta \sin \psi \cos \phi - \cos \psi \sin \phi \\ \cos \theta \cos \phi \end{bmatrix}$$

is the component of the orthogonal rotation matrix in the direction of $\mathbf{e}_z^e$; T is the total thrust force applied along $\mathbf{e}_z^b$; $\mathbf{I} = \text{diag}\{I_{xx}, I_{yy}, I_{zz}\}$ is a diagonal matrix of moments of inertia; $\boldsymbol{\tau} = [\tau_x \ \tau_y \ \tau_z]^\top$ is a vector of the rotation torques.

The thrust force T and the torque vector $\boldsymbol{\tau}$ act as the control inputs of the quadrotor system. They are all generated by the lifting forces T_1, T_2, T_3, T_4 produced by the four rotors, as presented below

$$\begin{bmatrix} T \\ \boldsymbol{\tau} \end{bmatrix} = \begin{bmatrix} l & l & l & l \\ 0 & -l & 0 & l \\ l & 0 & -l & 0 \\ -l & l & -l & l \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix} \quad (2)$$

where l is the distance from the rotors to the center of mass of the vehicle.

2.1.2. Input and state constraints

Given the physical limitation of the rotors, the inputs should comply with the following constraints:

$$T \in \{T \in \mathbb{R} \mid |T - mg| \leq \bar{T}\} \quad (3a)$$

$$\boldsymbol{\tau} \in \{\boldsymbol{\tau} \in \mathbb{R}^3 \mid \|\boldsymbol{\tau}\| \leq \bar{\tau}\} \quad (3b)$$

where $\bar{T}, \bar{\tau}$ are positive constants specifying acceptable changing ranges for T and τ .

On the other hand, acrobatic maneuvers of the vehicle should be avoided for the sake of flight safety. This requires the rotation angles should evolve within the following set

$$\mathbf{q} \in \Omega_q \triangleq \{\mathbf{q} \in \mathbb{R}^3 \mid \|\mathbf{q}\| \leq \bar{q}\} \quad (4)$$

where $\bar{q}$ is the upper bound of the Euclidean norm of $\mathbf{q}$.

2.1.3. Actuator faults

As the actuators of the quadrotor control system, rotors have the possibility of malfunctions such as low voltage supply or blade deflection. The impacts of common actuator failures can be modeled as either multiplicative or additive uncertainties in the output of the actuators as follows (Hu, Shao, & Guo, 2017):

$$\begin{bmatrix} T \\ \boldsymbol{\tau} \end{bmatrix} = \underbrace{\begin{bmatrix} -\sigma_T & \sigma_{\tau_x} & \sigma_{\tau_y} & \sigma_{\tau_z} \end{bmatrix}}_{\sigma_\tau} \begin{bmatrix} u_T \\ u_{\tau_x} \\ u_{\tau_y} \\ u_{\tau_z} \end{bmatrix} + \underbrace{\begin{bmatrix} \sigma_{T_0} \\ \sigma_{\tau_{x0}} \\ \sigma_{\tau_{y0}} \\ \sigma_{\tau_{z0}} \end{bmatrix}}_{\sigma_{\tau_0}} \quad (5)$$

where u_T and $\mathbf{u}_\tau = [u_{\tau_x} \ u_{\tau_y} \ u_{\tau_z}]^\top$ are the control commands to be determined later in Section 4; σ_T and $\sigma_{\tau_x, \tau_y, \tau_z}$ denote the

multiplicative losses of thrust and torques; σ_{T_0} and $\sigma_{\tau_{x0}, \tau_{y0}, \tau_{z0}}$ are additive parameters used to represent the bias caused by faults. In this work, no fault detection or identification device is equipped in the flight control system, so the real-time values of fault parameters $\sigma_T, \sigma_{T_0}, \sigma_\tau$ and σ_{τ_0} are unknown.

Assumption 1. The multiplicative parameters $\sigma_T, \sigma_{\tau_x, \tau_y, \tau_z}$ are bounded by an upper bound of 1 and a lower bound denoted as $\underline{\sigma} \in (0, 1)$. The unknown fault parameters σ_{T_0} and σ_{τ_0} are bounded above by $\bar{\sigma}_{T_0} \in \mathbb{R}^+$ and $\bar{\sigma}_{\tau_0} \in \mathbb{R}^+$ in their absolute value and Euclidean norm – that is $|\sigma_{T_0}| \leq \bar{\sigma}_{T_0}$ and $\|\sigma_{\tau_0}\| \leq \bar{\sigma}_{\tau_0}$. Also, $\sigma_T, \sigma_\tau, \sigma_{T_0}$ and σ_{τ_0} are differentiable and slowly varying so their time derivatives can be regarded as 0 in the performance evaluation without losing much of the analytical accuracy.

Remark 1. Even though the exact fault information, like the time of occurrence or the magnitude, is hard to obtain without fault detection mechanisms, it is practical to estimate a rough impact range of the faults from historical data. Therefore, the above assumption that the unknown fault parameters have known bounds is reasonable.

In order to guarantee the satisfaction of the input constraints (3) even in the most extreme cases of faults, the control commands u_T and $\mathbf{u}_\tau$ should be further restricted to the following tightened sets

$$u_T \in \Omega_T \triangleq \{u_T \in \mathbb{R} \mid |u_T - mg| \leq \bar{T} - \bar{\sigma}_{T_0}\} \quad (6a)$$

$$\mathbf{u}_\tau \in \Omega_\tau \triangleq \{\mathbf{u}_\tau \in \mathbb{R}^3 \mid \|\mathbf{u}_\tau\| \leq \bar{\tau} - \bar{\sigma}_{\tau_0}\} \quad (6b)$$

2.2. Tracking control objectives

This work aims to develop a control scheme for generating the control commands u_T and $\mathbf{u}_\tau$ to fulfill the following control objectives:

- (O1) Drive the quadrotor modeled by (1) to track a prespecified flight trajectory denoted as $\mathbf{p}^d = [x^d \ y^d \ z^d]^\top$;
- (O2) Stabilize the rotation of the quadrotor.

The control system is developed to achieve (O1) and (O2) in the presence of the input constraints (6), the state constraint (4), and the actuator faults (5).

Assumption 2. The specified desired trajectory $\mathbf{p}^d$ is slowly time-varying with respect to the quadrotor dynamics. $\mathbf{p}^d$ is piece-wise continuous with the outer-loop sampling period. Its first- and second-order derivatives exist, denoted as $\dot{\mathbf{p}}^d$ and $\ddot{\mathbf{p}}^d$.

3. Model decomposition

In the quadrotor model (1), (1a) represents the translational dynamics while (1b) represents the rotational dynamics. To enable the later control design with hierarchical architecture, model decomposition of (1a) and (1b) is required, which is legitimized by assuming that (1b) converges much faster than (1a). The difference in convergence rate, from another perspective, can be regarded as a separation in time-scales. Therefore, below we consider the problem in a dual-time-scale context.

For the fast-varying rotational dynamics, a “stretched” time variable $\check{t}$ is introduced by defining $\check{t} = t/\varepsilon$ with t representing the standard time scale and $\varepsilon \in (0, 1]$. The time derivatives of a system state vector $\mathbf{x}$ with respect to $\check{t}$ and t should meet the following scaling relationship

$$\frac{d\mathbf{x}}{d\check{t}} = \varepsilon \frac{d\mathbf{x}}{dt} \quad (7)$$

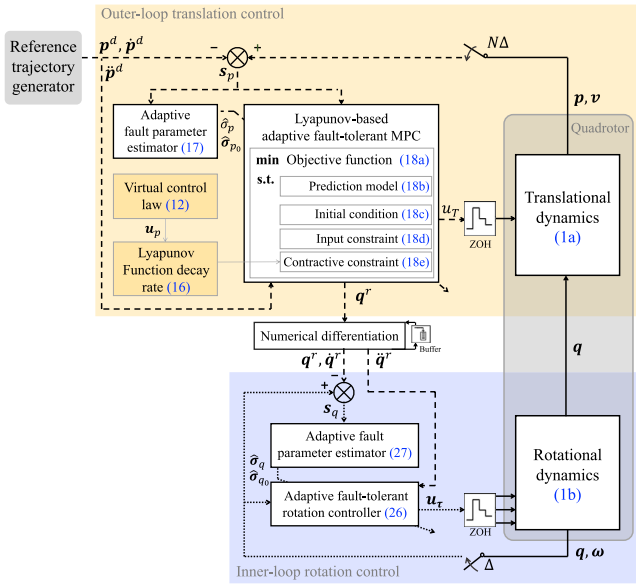


Fig. 2. Block diagram of the hierarchical control system.

Then, we introduce new notations of $\mathbf{q}$ and $\boldsymbol{\omega}$ in the fast time scale as $\tilde{\mathbf{q}} \triangleq \mathbf{q}$ and $\tilde{\boldsymbol{\omega}} \triangleq \boldsymbol{\omega}$. Thereupon, the quadrotor dynamics (1) can be rewritten in the following dual-time-scale format

$$\begin{cases} \dot{\mathbf{p}} = \mathbf{v} \\ m\dot{\mathbf{v}} = -m\mathbf{g} + \mathbf{r}(\mathbf{q})T \end{cases} \quad (8a)$$

$$\begin{cases} \varepsilon \dot{\tilde{\mathbf{q}}} = \tilde{\boldsymbol{\omega}} \\ \varepsilon \mathbf{I} \dot{\tilde{\boldsymbol{\omega}}} = -\tilde{\boldsymbol{\omega}} \times \mathbf{I} \tilde{\boldsymbol{\omega}} + \varepsilon^2 \boldsymbol{\tau} \end{cases} \quad (8b)$$

which is a standard singularly perturbed system with the perturbation parameter ε .

Model decomposition is produced by setting ε to 0. As $\varepsilon = 0$, transients of $\tilde{\mathbf{q}}$ and $\tilde{\boldsymbol{\omega}}$ diminish instantaneously. Assuming that the rotation controller is well designed to drive $\mathbf{q}$ to its reference signal, denoted as $\mathbf{q}^r$, $\mathbf{q}$ in the slower-varying translational subsystem (8a) can be approximated by $\mathbf{q}^r$. As a result, we obtain the following reduced-order translational dynamics

$$\begin{cases} \dot{\mathbf{p}} = \mathbf{v} \\ m\dot{\mathbf{v}} = -m\mathbf{g} + \mathbf{r}(\mathbf{q}^r)T \end{cases} \quad (9)$$

which is decoupled from the rotational dynamics (8b) evolving in the fast time-scale.

4. Control design

4.1. Dual-rate hierarchical architecture

Fig. 2 illustrates the block diagram of the developed hierarchical control system, in which an outer-loop translation control and an inner-loop rotation control are designed separately. The two control loops work in series: Firstly, the outer-loop translation control is developed to generate the required thrust force command u_T and $\mathbf{q}^r$ for steering the vehicle to perform certain prespecified translation behaviors; then, the inner-loop rotation control is developed to manipulate the actual attitude $\mathbf{q}$ to track $\mathbf{q}^r$ rapidly. The detailed design of the two control loops will be given later in Sections 4.2 and 4.3, respectively.

The developed inner-loop and outer-loop controllers are all implemented in a sample-and-hold fashion, meaning that the system is updated periodically at sampling instants and the computed control action is held until the next state measurement

is available. Given the expected dual-time-scale behavior of the translation and rotation control loops, it is natural to set different sampling rates for them: The sampling period for the fast-varying rotation control inner-loop is defined as Δ , and the outer-loop sampling period is $N\Delta$ with N being a positive integer. Under this dual-sampling-rate setting, the translation control is updated once while the rotation control is updated N times. The sequence of the sampling instants of the inner-loop is denoted as $\{k_i\Delta | k_i \in \{0, 1, 2, \dots\}\}$. Correspondingly, the sampling instant sequence of the outer-loop is $\{k_o\Delta | k_o \in \{0, N, 2N, \dots\}\}$.

In the subsequent design and analysis, we use notations with the time variable t in round brackets to represent signals evolving continuously, and use those with the sampling numbers k_i and k_o in square brackets to represent the measurements at discrete time instants. For example, in the rotation control loop, $\mathbf{q}[k_i]$ represents the measurement of the continuous state variable $\mathbf{q}(t)$ at the sampling instant $t = k_i\Delta$. In the translation control loop, $\mathbf{p}[k_o]$ is the measurement of $\mathbf{p}(t)$ at $t = k_o\Delta$.

4.2. Outer-loop translation control

4.2.1. Translational tracking error

The translation control, forming the outer-loop of the hierarchical control system, aims to realize tracking of the desired trajectory and to enforce a certain level of optimal performance subject to constraints and unknown actuator faults. Motivated by the translation tracking objectives, a sliding mode tracking error is introduced

$$\mathbf{s}_p = \lambda_p \mathbf{e}_p + \mathbf{e}_v \quad (10)$$

where $\mathbf{e}_p = \mathbf{p} - \mathbf{p}^d$ and $\mathbf{e}_v = \mathbf{v} - \dot{\mathbf{p}}^d$ are the position and velocity tracking errors. λ_p is chosen appropriately such that the polynomial $s + \lambda_p$ is Hurwitz. Then, $\mathbf{e}_p \rightarrow \mathbf{0}$ as $t \rightarrow 0$ on the sliding surface $\mathbf{s}_p = \mathbf{0}$. The translation tracking control objective (O1) can then be restated as

(O1') Keep the translational sliding mode tracking error $\mathbf{s}_p$ stay on, or near the sliding surface $\mathbf{s}_p = \mathbf{0}$.

Recalling the translational dynamics (9) and the actuator model (5), we have

$$\dot{\mathbf{s}}_p^- = \lambda_p \mathbf{e}_v - \mathbf{g} + \sigma_p^{-1} \mathbf{r}(\mathbf{q}^r) u_T + \sigma_{p_0} - \ddot{\mathbf{p}}^d \quad (11)$$

where the superscript $-$ implies that the derivative is obtained along the reduced translational dynamics (9); $\sigma_p = m/\sigma_T$; $\sigma_{p_0} = m^{-1} \mathbf{r}(\mathbf{q}^r) \sigma_{T_0}$.

Remark 2. From the boundedness of σ_T , σ_{T_0} and the fact that sine and cosine functions evolve within $[-1, 1]$, σ_p and σ_{p_0} in (11) also have bounded magnitudes and changing rates.

4.2.2. Auxiliary virtual control design

Before formulating the MPC problem, we need to firstly study the stability property achieved by a virtual control design that is developed to provide an auxiliary solution to stabilize the translational dynamics (9). We define a virtual control vector $\mathbf{u}_p$, the control law of which is designed as

$$\mathbf{u}_p \triangleq \mathbf{r}(\mathbf{q}^r) u_T = \hat{\sigma}_p (-c_p \mathbf{s}_p - \lambda_p \mathbf{e}_v + \mathbf{g} - \hat{\sigma}_{p_0} + \ddot{\mathbf{p}}^d) \quad (12)$$

where c_p is a strictly positive constant as the control gain of the virtual controller; $\hat{\sigma}_p$ and $\hat{\sigma}_{p_0}$ are online estimates of the unknown parameter σ_p and σ_{p_0} , updated by the following adaptive laws

$$\dot{\hat{\sigma}}_p = -\kappa_p \hat{\sigma}_p^{-1} \mathbf{s}_p^\top \mathbf{u}_p \quad (13a)$$

$$\dot{\hat{\sigma}}_{p_0} = \kappa_{p_0} \mathbf{s}_p \quad (13b)$$

with κ_p and κ_{p_0} being positive constants.

To characterize the closed-loop stability of the virtual control design, a Lyapunov function candidate is selected

$$V_p = \frac{1}{2} \|\mathbf{s}_p\|^2 + \frac{1}{2\sigma_p \kappa_p} e_{\sigma_p}^2 + \frac{1}{2\kappa_{p_0}} \|\mathbf{e}_{\sigma_{p_0}}\|^2 \quad (14)$$

where e_{σ_p} and $\mathbf{e}_{\sigma_{p_0}}$ are the adaptive estimation errors defined as $e_{\sigma_p} = \hat{\sigma}_p - \sigma_p$ and $\mathbf{e}_{\sigma_{p_0}} = \hat{\sigma}_{p_0} - \sigma_{p_0}$. Differentiating V_p along (11) yields

$$\begin{aligned} \dot{V}_p = & \mathbf{s}_p^\top \left(\lambda_p \mathbf{e}_v - \mathbf{g} + \frac{\hat{\sigma}_p}{\sigma_p} \hat{\sigma}_p^{-1} \mathbf{r}(\mathbf{q}^r) u_T + \sigma_{p_0} - \ddot{\mathbf{p}}^d \right) + \frac{e_{\sigma_p}}{\sigma_p \kappa_p} \dot{\sigma}_p \\ & + \frac{1}{\kappa_{p_0}} \dot{\sigma}_{p_0}^\top \mathbf{e}_{\sigma_{p_0}} \end{aligned} \quad (15)$$

Substituting (12) and (13) into it further delivers

$$\begin{aligned} \dot{V}_p = & \mathbf{s}_p^\top (\lambda_p \mathbf{e}_v - \mathbf{g} + \hat{\sigma}_p^{-1} \mathbf{r}(\mathbf{q}^r) u_T + \hat{\sigma}_{p_0} - \ddot{\mathbf{p}}^d) \\ & \leq -c_p \|\mathbf{s}_p\|^2 \end{aligned} \quad (16)$$

which explicitly characterizes the decay of V_p^- under the virtual controller $\mathbf{u}_p$.

4.2.3. Lyapunov-based fault-tolerant MPC

The developed Lyapunov-based fault-tolerant MPC framework consists of an adaptive online estimator and a Lyapunov-based MPC controller. The adaptive laws for estimating σ_p and σ_{p_0} within the interval $t \in [k_o \Delta, k_o \Delta + N \Delta]$ are designed as

$$\dot{\hat{\sigma}}_p(t) = -\kappa_p \hat{\sigma}_p^{-1} [k_o] u_T [k_o] \mathbf{s}_p [k_o]^\top \mathbf{r}(\mathbf{q}^r [k_o]) \quad (17a)$$

$$\dot{\hat{\sigma}}_{p_0}(t) = \kappa_{p_0} \mathbf{s}_p [k_o] \quad (17b)$$

where κ_p and κ_{p_0} are positive constants.

We now present the MPC problem formulation. To distinguish the model used for prediction from the real system model, we mark the internal variables of MPC with a check sign above. MPC determines $\mathbf{q}^r$ and u_T by solving a finite horizon optimization problem at each sampling instant. Taking the measured errors $\mathbf{e}_p[k_o]$ and $\mathbf{e}_v[k_o]$, the estimated parameters $\hat{\sigma}_p[k_o]$ and $\hat{\sigma}_{p_0}[k_o]$, and $\ddot{\mathbf{p}}^d$ over the prediction horizon as input elements, the MPC problem at k_o is constructed by

$$\min_{\check{\mathbf{q}}^r(\cdot), \check{u}_T(\cdot)} \int_{k_o}^{k_o+N_p N \Delta} \left(\|\check{\mathbf{s}}_p(t)\|_{\mathbf{Q}}^2 + \|\check{\mathbf{q}}^r(t)\|_{\mathbf{R}_q}^2 + |\check{u}_T(t) - mg|_{R_T}^2 \right) dt \quad (18a)$$

subject to

$$\begin{cases} \dot{\check{\mathbf{e}}}_p(t) = \check{\mathbf{e}}_v(t), \\ \dot{\check{\mathbf{e}}}_v(t) = -\mathbf{g} + \hat{\sigma}_p^{-1} [k_o] \mathbf{r}(\check{\mathbf{q}}^r(t)) \check{u}_T(t) + \hat{\sigma}_{p_0} [k_o] - \ddot{\mathbf{p}}^d(t), \end{cases} \quad \text{for } t \in [k_o \Delta, k_o \Delta + N_p N \Delta] \quad (18b)$$

$$\begin{cases} \check{\mathbf{e}}_p(t) = \mathbf{e}_p[k_o], \\ \check{\mathbf{e}}_v(t) = \mathbf{e}_v[k_o], \end{cases} \quad \text{for } t \in [k_o \Delta, k_o \Delta + N \Delta] \quad (18c)$$

$$\begin{cases} \check{\mathbf{q}}^r(\cdot) \in S_q(N \Delta) : [k_o \Delta, k_o \Delta + N_p N \Delta] \rightarrow \Omega_q \\ \check{u}_T(\cdot) \in S_T(N \Delta) : [k_o \Delta, k_o \Delta + N_p N \Delta] \rightarrow \Omega_T \end{cases} \quad (18d)$$

$$\begin{aligned} \check{\mathbf{s}}_p^\top(t) (\lambda_p \check{\mathbf{e}}_v(t) - \mathbf{g} + \hat{\sigma}_p^{-1} [k_o] \mathbf{r}(\check{\mathbf{q}}^r(t)) \check{u}_T(t) + \hat{\sigma}_{p_0} [k_o] - \ddot{\mathbf{p}}^d(t)) \\ \leq -c_p \|\check{\mathbf{s}}_p(t)\|^2, \end{aligned} \quad \text{for } t \in [k_o \Delta, k_o \Delta + N \Delta] \quad (18e)$$

where $\check{\mathbf{s}}_p = \lambda_p \check{\mathbf{e}}_p + \check{\mathbf{e}}_v$ is the predicted sling mode error; N_p is the number of samplings contained within the prediction horizon; $\mathbf{Q}$ and $\mathbf{R}_q$ are positive-definite and symmetric weighting matrices; R_T is a positive weighting scalar; $S_q(N \Delta)$ and $S_T(N \Delta)$ are families of piece-wise continuous functions with sampling period $N \Delta$, mapping $[k_o \Delta, k_o \Delta + N_p N \Delta]$ into Ω_q and Ω_T , respectively; $\check{\mathbf{q}}^r$ and $\check{u}_T$, belonging to $S_q(N \Delta)$ and $S_T(N \Delta)$, are control profiles defined over $[k_o \Delta, k_o \Delta + N_p N \Delta]$ and the decision variables of the

optimization problem; $\check{\mathbf{e}}_p$ and $\check{\mathbf{e}}_v$, also defined over $[k_o \Delta, k_o \Delta + N_p N \Delta]$, are predicted profiles of tracking errors driven by $\check{\mathbf{q}}^r$ and $\check{u}_T$.

In the optimization problem, (18b) is the prediction model to predict the future evolution of $\check{\mathbf{e}}_p$ and $\check{\mathbf{e}}_v$ with their initial values defined as (18c). In the prediction model, $\hat{\sigma}_p$ and $\hat{\sigma}_{p_0}$ are estimates of the unknown parameters updated by (17). (18d) ensures the input constraints. The construction of the contractive constraint (18e) is inspired by (16), enforcing a decay in the value of the Lyapunov function by at least the rate achieved by the virtual controller $\mathbf{u}_p$ during the first sampling interval of the prediction horizon.

The optimal solution of (18), denoted as $\check{\mathbf{q}}^{r*}$ and $\check{u}_T^*$, is implemented in a receding horizon manner. That is, the translation control commands at k_o equal the first element of $\check{\mathbf{q}}^{r*}$ and $\check{u}_T^*$ solved at k_o

$$\begin{cases} \mathbf{q}^r(t) = \check{\mathbf{q}}^{r*}(t; k_o) \\ u_T(t) = \check{u}_T^*(t; k_o) \end{cases} \quad (19)$$

where $t \in [k_o \Delta, k_o \Delta + N \Delta]$. This process is repeated and the translation control commands will be updated at the next outer-loop sampling instant $k_o \Delta + N \Delta$.

With the initial condition (18d) defined, the actual sliding mode error $\mathbf{s}_p[k_o]$ and the velocity error $\mathbf{e}_v[k_o]$ also obey the contractive constraint (18e). Thus, the following inequality holds:

$$\begin{aligned} \mathbf{s}_p^\top[k_o] (\lambda_p \mathbf{e}_v[k_o] - \mathbf{g} + \hat{\sigma}_p^{-1} [k_o] \mathbf{r}(\mathbf{q}^r[k_o]) u_T[k_o] + \hat{\sigma}_{p_0} [k_o] - \ddot{\mathbf{p}}^d[k_o]) \\ \leq -c_p \|\mathbf{s}_p[k_o]\|^2 \end{aligned} \quad (20)$$

which will be used later in Section 5 for stability analysis.

Lemma 1. There always exists a feasible solution for the optimization problem (18), which can be constructed as

$$\check{\mathbf{q}}^0 = \begin{cases} \mathbf{q}^0, & \text{for } t \in [k_o \Delta, k_o \Delta + N \Delta] \\ [0 \ 0 \ 0]^\top, & \text{for } t \in [k_o \Delta + N \Delta, k_o \Delta + N_p N \Delta] \end{cases} \quad (21a)$$

$$\check{u}_T^0 = \begin{cases} u_T^0, & \text{for } t \in [k_o \Delta, k_o \Delta + N \Delta] \\ mg, & \text{for } t \in [k_o \Delta + N \Delta, k_o \Delta + N_p N \Delta] \end{cases} \quad (21b)$$

where

$$\mathbf{q}^0 = \begin{bmatrix} \phi^0 \\ \theta^0 \\ \psi^0 \end{bmatrix} = \begin{bmatrix} \arctan \left(\frac{u_{p1} \sin \psi^0 - u_{p2} \cos \psi^0}{\sqrt{u_{p1}^2 + u_{p2}^2 + u_{p3}^2}} \right) \\ \arctan \left(\frac{u_{p1} \cos \psi^0 + u_{p2} \sin \psi^0}{u_{p3}} \right) \\ \psi^0 \end{bmatrix}$$

$$\begin{aligned} u_T^0 = & u_{p1} (\sin \theta^0 \cos \psi^0 \cos \phi^0 + \sin \psi^0 \sin \phi^0) + u_{p2} (\sin \theta^0 \sin \psi^0 \\ & \cos \phi^0 - \cos \psi^0 \sin \phi^0) + u_{p3} \cos \theta^0 \cos \phi^0 \end{aligned}$$

in which u_{p1} , u_{p2} , and u_{p3} are the first, second, and third elements of the virtual controller $\mathbf{u}_p$ developed in (12); ψ^0 could be any desired yaw angle satisfying (4).

Proof. For $t \in [k_o \Delta, k_o \Delta + N \Delta]$, the contractive constraint (18e) is satisfied by (21), since the virtual control law $\mathbf{u}_p$ satisfies (16). The input constraint (18d) is also satisfied by $\mathbf{q}^0$ and u_T^0 as long as the control gain c_p is chosen appropriately. For $t \in [k_o \Delta + N \Delta, k_o \Delta + N_p N \Delta]$, (18d) is not required to be satisfied while (18d) is automatically met. Therefore, we can prove that $\check{\mathbf{q}}^0$ and $\check{u}_T^0$ given in (21) provides feasible solutions for (18).

4.3. Inner-loop rotation control

We now turn to the inner-loop control design for the rotational subsystem (8b) in the fast time-scale. With $\mathbf{q}^r$ determined by the outer-loop translation control, the rotational control intends to drive $\mathbf{q}$ to track $\mathbf{q}^r$.

Similarly, a sliding mode tracking error of the rotation motion is defined in the fast time-scale

$$\check{\mathbf{s}}_q = \lambda_q \check{\mathbf{e}}_q + \check{\mathbf{e}}_\omega \quad (22)$$

where $\check{\mathbf{e}}_q = \check{\mathbf{q}} - \mathbf{q}^r$ and $\check{\mathbf{e}}_\omega = \check{\boldsymbol{\omega}} - \varepsilon \dot{\mathbf{q}}^r$; λ_q is chosen to be strictly positive. We also have $\check{\mathbf{e}}_q \rightarrow \mathbf{0}$ as $t \rightarrow 0$ when $\check{\mathbf{s}}_q = \mathbf{0}$. Thereafter, the rotation stabilization objective (O2) can be restated as

(O2') Keep $\check{\mathbf{s}}_q$ stay on, or near the sliding surface $\check{\mathbf{s}}_q = \mathbf{0}$.

Recalling the rotational dynamics (8b) and the actuator model (5), the dynamics of $\check{\mathbf{s}}_q$ is obtained as

$$\varepsilon \dot{\check{\mathbf{s}}}_q = \lambda_q \check{\mathbf{e}}_\omega + \mathbf{f}_q(\check{\boldsymbol{\omega}}) + \varepsilon^2 (\boldsymbol{\sigma}_q^{-1} \mathbf{u}_\tau + \boldsymbol{\sigma}_{q_0} - \ddot{\mathbf{q}}^r) \quad (23)$$

where $\mathbf{f}_q(\check{\boldsymbol{\omega}}) = -\mathbf{I}^{-1}(\check{\boldsymbol{\omega}} \times \mathbf{I} \check{\boldsymbol{\omega}})$; $\boldsymbol{\sigma}_q = \mathbf{I} \boldsymbol{\sigma}_\tau^{-1}$ and $\boldsymbol{\sigma}_{q_0} = \mathbf{I}^{-1} \boldsymbol{\sigma}_{\tau_0}$ are unknown parameters with known bounds. According to (23), an adaptive fault-tolerant rotation control law is developed as

$$\mathbf{u}_\tau = \hat{\boldsymbol{\sigma}}_q \left(\frac{1}{\varepsilon^2} (-c_q \check{\mathbf{s}}_q - \lambda_q \check{\mathbf{e}}_\omega - \mathbf{f}_q(\check{\boldsymbol{\omega}})) - \hat{\boldsymbol{\sigma}}_{q_0} + \ddot{\mathbf{q}}^r \right) \quad (24)$$

where c_q is strictly positive; $\hat{\boldsymbol{\sigma}}_q = \text{diag}\{\hat{\sigma}_{q_x}, \hat{\sigma}_{q_y}, \hat{\sigma}_{q_z}\}$ and $\hat{\boldsymbol{\sigma}}_{q_0} = [\hat{\sigma}_{q_{x0}} \ \hat{\sigma}_{q_{y0}} \ \hat{\sigma}_{q_{z0}}]^\top$ are estimates of $\boldsymbol{\sigma}_q$ and $\boldsymbol{\sigma}_{q_0}$, updated by the following adaptive laws

$$\dot{\hat{\sigma}}_{q_x} = -\varepsilon \kappa_q \check{\mathbf{s}}_q^\top \text{diag}\{\hat{\sigma}_{q_x}^{-1}, 0, 0\} \mathbf{u}_\tau \quad (25a)$$

$$\dot{\hat{\sigma}}_{q_y} = -\varepsilon \kappa_q \check{\mathbf{s}}_q^\top \text{diag}\{0, \hat{\sigma}_{q_y}^{-1}, 0\} \mathbf{u}_\tau \quad (25b)$$

$$\dot{\hat{\sigma}}_{q_z} = -\varepsilon \kappa_q \check{\mathbf{s}}_q^\top \text{diag}\{0, 0, \hat{\sigma}_{q_z}^{-1}\} \mathbf{u}_\tau \quad (25c)$$

$$\dot{\hat{\boldsymbol{\sigma}}}_{q_0} = \varepsilon \kappa_{q_0} \check{\mathbf{s}}_q \quad (25d)$$

where κ_q and κ_{q_0} are positive constants.

Under the sample-and-hold implementation setup, the rotation control law is updated discretely. The updated rotation control law for $t \in [k_i \Delta, k_i \Delta + \Delta)$ is

$$\mathbf{u}_\tau(t) = \hat{\boldsymbol{\sigma}}_q[k_i] \left(\frac{1}{\varepsilon^2} (-c_q \check{\mathbf{s}}_q[k_i] - \lambda_q \check{\mathbf{e}}_\omega[k_i] - \mathbf{f}_q(\check{\boldsymbol{\omega}}[k_i])) - \hat{\boldsymbol{\sigma}}_{q_0}[k_i] + \ddot{\mathbf{q}}^r[k_i] \right) \quad (26)$$

The sampled-data adaptive laws for $t \in [k_i \Delta, k_i \Delta + \Delta)$ are

$$\dot{\hat{\sigma}}_{q_x}(t) = -\varepsilon \kappa_q \check{\mathbf{s}}_q^\top[k_i] \text{diag}\{\hat{\sigma}_{q_x}^{-1}[k_i], 0, 0\} \mathbf{u}_\tau[k_i] \quad (27a)$$

$$\dot{\hat{\sigma}}_{q_y}(t) = -\varepsilon \kappa_q \check{\mathbf{s}}_q^\top[k_i] \text{diag}\{0, \hat{\sigma}_{q_y}^{-1}[k_i], 0\} \mathbf{u}_\tau[k_i] \quad (27b)$$

$$\dot{\hat{\sigma}}_{q_z}(t) = -\varepsilon \kappa_q \check{\mathbf{s}}_q^\top[k_i] \text{diag}\{0, 0, \hat{\sigma}_{q_z}^{-1}[k_i]\} \mathbf{u}_\tau[k_i] \quad (27c)$$

$$\dot{\hat{\boldsymbol{\sigma}}}_{q_0}(t) = \varepsilon \kappa_{q_0} \check{\mathbf{s}}_q[k_i] \quad (27d)$$

Remark 3. Higher-order time derivatives of $\mathbf{q}^r$ are required for generating the rotation control signal. $\dot{\mathbf{q}}^r$ and $\ddot{\mathbf{q}}^r$ can be approximated by doing numerical differentiation with $\mathbf{q}^r$ and its historical data, as is done in Pérez-Alcocer et al. (2016).

If presented under the standard time-scale of the outer-loop, the counter part of $\check{\mathbf{s}}_q$ is

$$\mathbf{s}_q = \frac{\check{\mathbf{s}}_q}{\varepsilon} = \frac{\lambda_q}{\varepsilon} \mathbf{e}_q + \mathbf{e}_\omega \quad (28)$$

where $\mathbf{e}_q = \mathbf{q} - \mathbf{q}^r$ and $\mathbf{e}_\omega = \boldsymbol{\omega} - \dot{\mathbf{q}}^r$. Following that, the rotation control law (24) can be expressed in the outer-loop time-scale as

$$\mathbf{u}_\tau = \hat{\boldsymbol{\sigma}}_q \left(-\frac{k_q}{\varepsilon} \mathbf{s}_q - \frac{\lambda_q}{\varepsilon} \mathbf{e}_\omega - \mathbf{f}_q(\boldsymbol{\omega}) - \hat{\boldsymbol{\sigma}}_{q_0} + \ddot{\mathbf{q}}^r \right) \quad (29)$$

Remark 4. Expression (29) explicitly indicates that the singular perturbation parameter ε formalizes the high-gain property of the

rotational control law — the smaller ε is set to be, the higher the control gain is. Therefore, the maximal admissible value of ε , which will be given later in Section 5, can quantify exactly how “high” the rotation control gain should be in view of the overall stability.

5. Closed-loop stability analysis

The outer-loop and inner-loop controllers are designed based on the decomposed model (9) and (8b) without considering the interaction of the translational and rotational dynamics. In this section, we will evaluate the overall closed-loop stability of the full dynamics with the disregarded interaction taken back into consideration. Furthermore, the uncontrolled inter-sample behaviors of the sample-data system are considered.

We take the following steps to analyze the stability. Firstly, Lyapunov functions for the translational dynamics and rotational dynamics are constructed respectively, and their derivatives with the developed translation and rotation control schemes are evaluated. After that, a composite Lyapunov function is formed by adding the two Lyapunov functions together, whose derivative is then proven to be upper bounded by a negative constant under certain stability conditions.

5.1. Lyapunov function for the translational dynamics

We start from the Lyapunov function V_p defined previously in (14). Since the control is implemented in a sample-and-hold setup, it is natural to look at the derivative of V_p between two adjacent sampling instants. For $t \in [k_i \Delta, k_i \Delta + \Delta)$, differentiating V_p along the full-order translational dynamics (1a) gives

$$\dot{V}_p(t) = \dot{V}_p^-(t) + D(t) \quad (30)$$

where $\dot{V}_p^-(t)$ is the derivative of V_p along the reduced translational dynamics (9) and the adaptive updating laws (17); $D(t)$ describes the deviation between $\dot{V}_p$ and $\dot{V}_p^-$, resulting from the neglected transient dynamics of $\mathbf{q}$ towards $\mathbf{q}^r$, given as follows

$$\begin{aligned} \dot{V}_p^-(t) &= \mathbf{s}_p^\top(t) (\lambda_p \mathbf{e}_v(t) - \mathbf{g} + \sigma_p^{-1}(t) \mathbf{r}(\mathbf{q}^r[k_o]) u_T[k_o] + \sigma_{p_0}(t) - \ddot{\mathbf{p}}^d(t)) \\ &\quad - \frac{\mathbf{e}_{\sigma_p}(t)}{\sigma_p(t)} \hat{\sigma}_p^{-1}[k_o] \mathbf{s}_p^\top[k_o] \mathbf{r}(\mathbf{q}^r[k_o]) u_T[k_o] + \mathbf{s}_p^\top[k_o] \mathbf{e}_{\sigma_{p_0}}[k_o] \\ D(t) &= \mathbf{s}_p^\top(t) \sigma_p^{-1}(t) (\mathbf{r}(\mathbf{q}(t)) - \mathbf{r}(\mathbf{q}^r[k_o])) (u_T[k_o] + \sigma_{T_0}(t)) \end{aligned}$$

in which, $k_o = N \lfloor k_i / N \rfloor$ denotes the latest outer-loop sampling instant in or before the considered time interval $[k_i \Delta, k_i \Delta + \Delta)$. Eq. (30) can be further written as

$$\dot{V}_p(t) = \dot{V}_p^-[k_o] + D[k_o] + (\dot{V}_p^-(t) - \dot{V}_p^-[k_o]) + (D(t) - D[k_o]) \quad (31)$$

where

$$\begin{aligned} \dot{V}_p^-[k_o] &= \mathbf{s}_p^\top[k_o] (\lambda_p \mathbf{e}_v[k_o] - \mathbf{g} + \sigma_p^{-1}[k_o] \mathbf{r}(\mathbf{q}^r[k_o]) u_T[k_o] - \ddot{\mathbf{p}}^d[k_o]) \\ &\quad - \frac{\mathbf{e}_{\sigma_p}[k_o]}{\sigma_p[k_o]} \hat{\sigma}_p^{-1}[k_o] \mathbf{s}_p^\top[k_o] \mathbf{r}(\mathbf{q}^r[k_o]) u_T[k_o] + \mathbf{s}_p^\top[k_o] \mathbf{e}_{\sigma_{p_0}}[k_o] \\ D[k_o] &= \mathbf{s}_p^\top[k_o] \sigma_p^{-1}[k_o] (\mathbf{r}(\mathbf{q}[k_o]) - \mathbf{r}(\mathbf{q}^r[k_o])) (u_T[k_o] + \sigma_{T_0}[k_o]) \end{aligned}$$

Substituting (20) to the right-hand side of $\dot{V}_p^-[k_o]$ yields

$$\begin{aligned} \dot{V}_p^-[k_o] &= \mathbf{s}_p^\top[k_o] (\lambda_p \mathbf{e}_v[k_o] - \mathbf{g} + \hat{\sigma}_p^{-1}[k_o] \mathbf{r}(\mathbf{q}^r[k_o]) u_T[k_o] + \hat{\boldsymbol{\sigma}}_{p_0}[k_o] \\ &\quad - \ddot{\mathbf{p}}^d[k_o]) \\ &\leq -c_p \|\mathbf{s}_p[k_o]\|^2 \end{aligned} \quad (32)$$

Then, we denote the stability regions for the sliding mode tracking errors $\mathbf{s}_p$ and $\check{\mathbf{s}}_q$ as $\Omega_{\rho_p} = \{\mathbf{s}_p \mid \|\mathbf{s}_p\|^2 \leq \rho_p\}$ and $\Omega_{\rho_q} = \{\check{\mathbf{s}}_q \mid \|\check{\mathbf{s}}_q\|^2 \leq \rho_q\}$. By the Lipschitz continuity of $\mathbf{s}_p$, $\check{\mathbf{s}}_q$, σ_p , σ_{p_0} , $\hat{\sigma}_p$, $\hat{\boldsymbol{\sigma}}_{p_0}$, the boundedness of σ_p and σ_{p_0} , and $\dot{\sigma}_p \approx 0$, $\dot{\sigma}_{p_0} \approx \mathbf{0}$, for all

$\mathbf{s}_p \in \Omega_{\rho_p}$, $\check{\mathbf{s}}_q \in \Omega_{\rho_q}$, $u_T \in \Omega_T$, we have

$$\begin{aligned} |\dot{V}_p^-(t) - \dot{V}_p^-[k_o]| &\leq L_{s_p} \|\mathbf{s}_p(t) - \mathbf{s}_p[k_o]\| + L_{\hat{s}_{p0}} \|\hat{\mathbf{s}}_{p0}(t) - \hat{\mathbf{s}}_{p0}[k_o]\| \\ &\quad + L_{\hat{s}_p} \|\hat{s}_p(t) - \hat{s}_p[k_o]\| \end{aligned} \quad (33)$$

$$|D(t) - D[k_o[k_o]]| \leq L_{s_p}^D \|\mathbf{s}_p(t) - \mathbf{s}_p[k_o]\| + L_{s_q}^D \|\mathbf{s}_q(t) - \mathbf{s}_q[k_o]\| \quad (34)$$

and

$$\|\mathbf{s}_p(t) - \mathbf{s}_p[k_o]\| \leq M_{s_p} N \Delta \quad (35a)$$

$$\|\mathbf{s}_q(t) - \mathbf{s}_q[k_o]\| \leq M_{s_q} N \Delta \quad (35b)$$

$$|\hat{s}_p(t) - \hat{s}_p[k_o]| \leq M_{\hat{s}_p} N \Delta \quad (35c)$$

$$\|\hat{\mathbf{s}}_{p0}(t) - \hat{\mathbf{s}}_{p0}[k_o]\| \leq M_{\hat{s}_{p0}} N \Delta \quad (35d)$$

where L_{s_p} , $L_{\hat{s}_{p0}}$, $L_{\hat{s}_p}$, $L_{s_p}^D$, $L_{s_q}^D$, M_{s_p} , M_{s_q} , $M_{\hat{s}_{p0}}$ and $M_{\hat{s}_p}$ are positive Lipschitz constants that can be obtained by evaluating the bounds of dynamics within the stability regions and the input constraints.

Since $\left\| \frac{\partial \mathbf{r}(\mathbf{q})}{\partial \mathbf{q}} \right\| = 1$, we can obtain

$$\begin{aligned} D[k_o] &\leq \left(\frac{\bar{T}}{m} + g \right) \|\mathbf{s}_p[k_o]\| \|\mathbf{r}(\mathbf{q}[k_o]) - \mathbf{r}(\mathbf{q}^r[k_o])\| \\ &\leq \left(\frac{\bar{T}}{m} + g \right) \|\mathbf{s}_p[k_o]\| \|\mathbf{e}_q[k_o]\| \\ &\leq \frac{\bar{T} + mg}{m\sqrt{2\lambda_q}} \|\mathbf{s}_p[k_o]\| \|\check{\mathbf{s}}_q[k_o]\| \end{aligned} \quad (36)$$

By substituting (32)–(36) together into (31), we have

$$\begin{aligned} \dot{V}_p(t) &\leq \dot{V}_p^-[k_o] + D[k_o] + |\dot{V}_p^-(t) - \dot{V}_p^-[k_o]| + |D(t) - D[k_o]| \\ &\leq -c_p \|\mathbf{s}_p[k_o]\|^2 + \alpha \|\mathbf{s}_p[k_o]\| \|\check{\mathbf{s}}_q[k_o]\| + \gamma_p N \Delta \end{aligned} \quad (37)$$

where $\gamma_p = (L_{s_p} + L_{s_p}^D)M_{s_p} + L_{s_q}^D M_{s_q} + L_{\hat{s}_{p0}} M_{\hat{s}_{p0}} + L_{\hat{s}_p} M_{\hat{s}_p}$ and $\alpha = \frac{\bar{T} + mg}{m\sqrt{2\lambda_q}}$. Since $k_o = N \lfloor k_i/N \rfloor$, we also have

$$\|\mathbf{s}_p[k_i]\|^2 - \|\mathbf{s}_p[k_o]\|^2 \leq 2\sqrt{\rho_p} M_{s_p} (N-1) \Delta \quad (38)$$

$$\begin{aligned} \|\mathbf{s}_p[k_o]\| \|\check{\mathbf{s}}_q[k_o]\| - \|\mathbf{s}_p[k_i]\| \|\check{\mathbf{s}}_q[k_i]\| &\leq \sqrt{\rho_q} M_{s_p} (N-1) \Delta \\ &\quad + \sqrt{\rho_p} M_{s_q} (N-1) \Delta \end{aligned} \quad (39)$$

Finally, the upper bound of $\dot{V}_p(t)$ for $t \in [k_i\Delta, (k_i+1)\Delta)$ is

$$\begin{aligned} \dot{V}_p(t) &\leq -c_p \|\mathbf{s}_p[k_i]\|^2 + \alpha \|\mathbf{s}_p[k_i]\| \|\check{\mathbf{s}}_q[k_i]\| + \gamma_p N \Delta + \zeta (N-1) \Delta \\ &\quad + c_p \eta (N-1) \Delta \end{aligned} \quad (40)$$

where $\zeta = \alpha(\sqrt{\rho_q} M_{s_p} + \sqrt{\rho_p} M_{s_q})$ and $\eta = 2\sqrt{\rho_p} M_{s_p}$.

5.2. Lyapunov function for the rotational dynamics

We now move on to the closed-loop rotational dynamics. Firstly, choose a candidate for the Lyapunov function

$$V_q = \frac{1}{2} \|\check{\mathbf{s}}_q\|^2 + \frac{1}{2\sigma_{q_x} \kappa_q} e_{\alpha_{q_x}}^2 + \frac{1}{2\sigma_{q_y} \kappa_q} e_{\alpha_{q_y}}^2 + \frac{1}{2\sigma_{q_z} \kappa_q} e_{\alpha_{q_z}}^2 + \frac{1}{2\kappa_{q_0}} \|\mathbf{e}_{\alpha_{q_0}}\|^2 \quad (41)$$

Differentiating V_q along the sliding mode error dynamics (23) and substituting the adaptive laws (27) over the sampling interval

$[k_i\Delta, (k_i+1)\Delta)$ give

$$\begin{aligned} \dot{V}_q(t) &= \frac{1}{\varepsilon} \check{\mathbf{s}}_q^\top(t) (\lambda_q \check{\mathbf{e}}_\omega(t) + \mathbf{f}_q(\check{\omega}(t)) + \varepsilon^2 (\sigma_q^{-1}(t) \mathbf{u}_\tau[k_i] + \sigma_{q_0}(t) \\ &\quad - \check{\mathbf{q}}^r(t))) - \varepsilon \check{\mathbf{s}}_q^\top[k_i] \sigma_q^{-1}(t) \mathbf{e}_{\sigma_q}(t) \hat{\sigma}_q^{-1}[k_i] \mathbf{u}_\tau[k_i] \\ &\quad + \varepsilon \check{\mathbf{s}}_q^\top[k_i] \mathbf{e}_{\sigma_{q_0}}(t) \end{aligned} \quad (42)$$

which can be rewritten as

$$\dot{V}_q(t) = \dot{V}_q[k_i] + (\dot{V}_q(t) - \dot{V}_q[k_i]) \quad (43)$$

where

$$\begin{aligned} \dot{V}_q[k_i] &= \frac{1}{\varepsilon} \check{\mathbf{s}}_q^\top[k_i] (\lambda_q \check{\mathbf{e}}_\omega[k_i] + \mathbf{f}_q(\check{\omega}[k_i]) + \varepsilon^2 (\sigma_q^{-1}[k_i] \mathbf{u}_\tau[k_i] + \sigma_{q_0}[k_i] \\ &\quad - \check{\mathbf{q}}^r[k_i])) - \varepsilon \check{\mathbf{s}}_q^\top[k_i] \sigma_q^{-1}[k_i] \mathbf{e}_{\sigma_q}[k_i] \hat{\sigma}_q^{-1}[k_i] \mathbf{u}_\tau[k_i] \\ &\quad + \varepsilon \check{\mathbf{s}}_q^\top[k_i] \mathbf{e}_{\sigma_{q_0}}[k_i] \end{aligned}$$

Substituting the rotation control law (29) to $\dot{V}_q[k_i]$, we have

$$\begin{aligned} \dot{V}_q[k_i] &= \frac{1}{\varepsilon} \check{\mathbf{s}}_q^\top[k_i] (\lambda_q \check{\mathbf{e}}_\omega[k_i] + \mathbf{f}_q(\check{\omega}[k_i]) + \varepsilon^2 (\hat{\sigma}_q^{-1}[k_i] \mathbf{u}_\tau[k_i] + \hat{\sigma}_{q_0}[k_i] \\ &\quad - \check{\mathbf{q}}^r[k_i])) \\ &= -\frac{c_q}{\varepsilon} \|\check{\mathbf{s}}_q[k_i]\|^2 \end{aligned} \quad (44)$$

Similarly, according to the Lipschitz continuity, the boundedness of σ_q , σ_{q_0} , $\hat{\sigma}_q$, $\hat{\sigma}_{q_0}$, and $\hat{\sigma}_q \approx \mathbf{0}$, $\hat{\sigma}_{q_0} \approx \mathbf{0}$, we have

$$\begin{aligned} |\dot{V}_q^-(t) - \dot{V}_q^-[k_i]| &\leq L_{s_q} \|\check{\mathbf{s}}_q(t) - \check{\mathbf{s}}_q[k_i]\| + L_{\hat{s}_{q0}} \|\hat{\mathbf{s}}_{q0}(t) - \hat{\mathbf{s}}_{q0}[k_i]\| \\ &\quad + L_{\hat{s}_q} \|\hat{s}_q(t) - \hat{s}_q[k_i]\| \end{aligned} \quad (45)$$

and

$$\|\check{\mathbf{s}}_q(t) - \check{\mathbf{s}}_q[k_i]\| \leq M_{s_q} \Delta \quad (46a)$$

$$\|\hat{\mathbf{s}}_{q0}(t) - \hat{\mathbf{s}}_{q0}[k_i]\| \leq M_{\hat{s}_{q0}} \Delta \quad (46b)$$

$$|\hat{s}_q(t) - \hat{s}_q[k_i]| \leq M_{\hat{s}_q} \Delta \quad (46c)$$

where L_{s_q} , $L_{\hat{s}_{q0}}$, $L_{\hat{s}_q}$, $M_{\hat{s}_{q0}}$ and $M_{\hat{s}_q}$ are also positive Lipschitz constants.

Substituting (44)–(46) to (43) delivers

$$\dot{V}_q(t) \leq -\frac{c_q}{\varepsilon} \|\check{\mathbf{s}}_q[k_i]\|^2 + \gamma_q \Delta \quad (47)$$

where $\gamma_q = L_{s_q} M_{s_q} + L_{\hat{s}_{q0}} M_{\hat{s}_{q0}} + L_{\hat{s}_q} M_{\hat{s}_q}$.

5.3. Composite Lyapunov function

By adding V_p and V_q up, we can construct a composite Lyapunov function to be the Lyapunov function candidate for the full quadrotor dynamics

$$\begin{aligned} V &= V_p + V_q \\ &= \frac{1}{2} \|\mathbf{s}\|^2 + \frac{1}{2\sigma_{p_x} \kappa_p} e_{\alpha_{p_x}}^2 + \frac{1}{2\kappa_{p_0}} \|\mathbf{e}_{\alpha_{p_0}}\|^2 + \frac{1}{2\sigma_{q_x} \kappa_q} e_{\alpha_{q_x}}^2 + \frac{1}{2\sigma_{q_y} \kappa_q} e_{\alpha_{q_y}}^2 \\ &\quad + \frac{1}{2\sigma_{q_z} \kappa_q} e_{\alpha_{q_z}}^2 + \frac{1}{2\kappa_{q_0}} \|\mathbf{e}_{\alpha_{q_0}}\|^2 \end{aligned} \quad (48)$$

where $\mathbf{s} = [\mathbf{s}_p^\top \check{\mathbf{s}}_q^\top]^\top$ can be regarded as the full-state sliding mode tracking error.

The composite Lyapunov function V is in the positive-definite quadratic form of the tracking and estimation errors, so it can be upper bounded by

$$V \leq \frac{1}{2} \left\| \begin{bmatrix} \|\mathbf{s}_p\| \\ \|\check{\mathbf{s}}_q\| \end{bmatrix} \right\|^2 + \mu \quad (49)$$

where $\mu = \frac{1}{2\kappa_{p0}} \rho_{s_p} + \frac{3}{2\kappa_{q0} \lambda_{\min}(I)} \rho_{s_q} + \frac{1}{2\kappa_{p0}} \rho_{s_{p0}} + \frac{1}{2\kappa_{q0}} \rho_{s_{q0}}$, in which ρ_{s_p} , ρ_{s_q} , $\rho_{s_{p0}}$ and $\rho_{s_{q0}}$ are used to define the stability regions

of the estimation errors as $\Omega_{\rho_p} = \{e_{\sigma_p} | |e_{\sigma_p}|^2 \leq \rho_p\}$, $\Omega_{\rho_q} = \{e_{\sigma_q} | |e_{\sigma_q}|^2 \leq \rho_q\}$, $\Omega_{\rho_{p_0}} = \{e_{\sigma_{p_0}} | \|e_{\sigma_{p_0}}\|^2 \leq \rho_{p_0}\}$, $\Omega_{\rho_{q_0}} = \{e_{\sigma_{q_0}} | \|e_{\sigma_{q_0}}\|^2 \leq \rho_{q_0}\}$.

5.4. Stability proof

With the composite Lyapunov function, we can prove the closed-loop stability of the proposed dual-loop and dual-rate hierarchical quadrotor control system using Lyapunov stability theory. **Theorem 1** summarizes the conclusion that the closed-loop tracking errors are always bounded and ultimately converge to a small region around the origin with sufficient stability conditions.

Theorem 1. Consider the closed-loop quadrotor dynamics (1a) and (1b), where the outer-loop controlled by the fault-tolerant MPC (18) with the adaptive law (17) and the inner-loop controlled by the rotation control law (26) with the adaptive estimating law (27). Given $\rho_s \in \mathbb{R}^+$ and $\epsilon \in \mathbb{R}^+$, if the selection of the inner-loop sampling period Δ , the outer-loop and inner-loop sampling period ratio N , and the perturbation parameter ϵ satisfy the following conditions

$$(C1) \quad \frac{1}{\epsilon} c_p c_q > \frac{\alpha^2}{4},$$

$$(C2) \quad \frac{\epsilon}{\Delta} + \Delta (\gamma_p N + \zeta(N-1) + \gamma_q) \leq 2\lambda_{\min}(\mathbf{C})(\rho_s - \mu) - c_p \eta(N-1)\Delta,$$

$\mathbf{s}_p(0) \in \Omega_{\rho_p}$, $\check{\mathbf{s}}_q(0) \in \Omega_{\rho_q}$, $e_{\sigma_p}(0) \in \Omega_{\rho_p}$, $e_{\sigma_{q,x,y,z}}(0) \in \Omega_{\rho_{q_0}}$, $e_{\sigma_{p_0}}(0) \in \Omega_{\rho_{p_0}}$, $e_{\sigma_{q_0}}(0) \in \Omega_{\rho_{q_0}}$, then $\mathbf{s}_p(t) \in \Omega_{\rho_p}$, $\check{\mathbf{s}}_q(t) \in \Omega_{\rho_q}$, $e_{\sigma_p}(t) \in \Omega_{\rho_p}$, $e_{\sigma_{q,x,y,z}}(t) \in \Omega_{\rho_{q_0}}$, $e_{\sigma_{p_0}}(t) \in \Omega_{\rho_{p_0}}$, $e_{\sigma_{q_0}}(t) \in \Omega_{\rho_{q_0}}$ for all $t > 0$, and

$$\limsup_{t \rightarrow \infty} \left\| \begin{bmatrix} \mathbf{s}_p^\top & \check{\mathbf{s}}_q^\top \end{bmatrix}^\top \right\| \leq \sqrt{\rho_{\min}} \quad (50)$$

$$\text{with } \rho_{\min} = \max \left\{ \left\| \begin{bmatrix} \mathbf{s}_p^\top(t+\Delta) & \check{\mathbf{s}}_q^\top(t+\Delta) \end{bmatrix}^\top \right\|^2 \mid V(t) \leq \rho_s \right\}.$$

Proof. The proof of Theorem 1 consists of two parts. We first prove that under the developed control design, the composite tracking error of the quadrotor dynamics will converge to a small set Ω_{ρ_s} after a finite number of sampling periods. Then, the closed-loop tracking error is ultimately bounded in the terminal set $\Omega_{\rho_{\min}}$ around the origin.

Part 1: Following the obtained upper bounds of V_p and V_q given by the right hand sides of (40) and (47), the time derivative of the composite Lyapunov function for all $t \in [k_i\Delta, (k_i+1)\Delta)$ is derived if $\mathbf{s}_p(0) \in \Omega_{\rho_p}$ and $\check{\mathbf{s}}_q(0) \in \Omega_{\rho_q}$, as given below

$$\dot{V}(t) \leq - \left[\|\mathbf{s}_p[k_i]\| \quad \|\check{\mathbf{s}}_q[k_i]\| \right] \mathbf{C} \begin{bmatrix} \|\mathbf{s}_p[k_i]\| \\ \|\check{\mathbf{s}}_q[k_i]\| \end{bmatrix} + \gamma_p N \Delta + \zeta(N-1)\Delta + c_p \eta(N-1)\Delta + \gamma_q \Delta \quad (51)$$

where $\mathbf{C} = \begin{bmatrix} c_p & -\frac{\alpha}{2} \\ -\frac{\alpha}{2} & \frac{1}{\epsilon} c_q \end{bmatrix}$. It is positive definite if condition (C1) holds. In this case, we have

$$\dot{V}(t) \leq - \lambda_{\min}(\mathbf{C}) \left\| \begin{bmatrix} \|\mathbf{s}_p[k_i]\| \\ \|\check{\mathbf{s}}_q[k_i]\| \end{bmatrix} \right\|^2 + \gamma_p N \Delta + \zeta(N-1)\Delta + c_p \eta(N-1)\Delta + \gamma_q \Delta \quad (52)$$

By recalling the upper bound of V given in (49), we have $\left\| \begin{bmatrix} \|\mathbf{s}_p\| \\ \|\check{\mathbf{s}}_q\| \end{bmatrix} \right\|^2 \geq 2(V - \mu)$. Then, if condition (C2) is satisfied and $\begin{bmatrix} \mathbf{s}_p[k_i] \\ \check{\mathbf{s}}_q[k_i] \end{bmatrix} \in \left\{ \begin{bmatrix} \mathbf{s}_p[k_i] \\ \check{\mathbf{s}}_q[k_i] \end{bmatrix} \mid \rho_s \leq V \leq \frac{1}{2}\rho_p + \frac{1}{2}\rho_q + \mu \right\}$, then the following inequality holds for all $t \in [k_i, (k_i+1))$

$$\dot{V}(t) \leq -\frac{\epsilon}{\Delta} \quad (53)$$

Table 1

Physical parameters of the quadrotor in simulations.

Parameters	Values
l – distance from rotors to the center	0.30486 (m)
m – mass of the quadrotor	2.618 (kg)
g – gravitational acceleration	9.81 (m/s ²)
I_{xx} – moment of inertia around $\mathbf{e}_x^e$	0.043467 (kg m ²)
I_{yy} – moment of inertia around $\mathbf{e}_y^e$	0.043467 (kg m ²)
I_{zz} – moment of inertia around $\mathbf{e}_z^e$	0.063267 (kg m ²)

By integrating it over $[k_i\Delta, (k_i+1)\Delta)$, we obtain $V(t) \leq V[k_i]$ for $t \in [k_i\Delta, (k_i+1)\Delta)$ and $V[k_i+1] \leq V[k_i] - \epsilon$. Therefore, it can be proven that, the full-state tracking error vector $\begin{bmatrix} \mathbf{s}_p^\top & \check{\mathbf{s}}_q^\top \end{bmatrix}^\top$ converges to Ω_{ρ_s} in a finite number samplings without leaving the stability region. The set Ω_{ρ_s} is a level set of the Lyapunov function V , defined as $\Omega_{\rho_s} = \left\{ \begin{bmatrix} \mathbf{s}_p^\top & \check{\mathbf{s}}_q^\top \end{bmatrix}^\top \mid V \leq \rho_s \right\}$.

Part 2: Because of the definition of $\rho_{\min}$, it can be concluded that once the composite tracking error $\begin{bmatrix} \mathbf{s}_p^\top & \check{\mathbf{s}}_q^\top \end{bmatrix}^\top$ enters Ω_{ρ_s} , then it will remain inside $\Omega_{\rho_{\min}} = \left\{ \begin{bmatrix} \mathbf{s}_p^\top & \check{\mathbf{s}}_q^\top \end{bmatrix}^\top \mid V \leq \rho_{\min} \right\}$ for all time. Finally, we obtain (50) and complete the proof of **Theorem 1**.

From the obtained stability conditions (C1) and (C2), it can be concluded that, given any $\rho_s \in \mathbb{R}^+$ (the size of the region that the tracking errors are expected to converge to) and any $\epsilon \in \mathbb{R}^+$ (the desired convergence step size), we can find appropriate combinations of sampling periods Δ and $N\Delta$, control gains c_p and c_q and singular perturbation parameters ϵ to guarantee the convergence of $\mathbf{s}_p$ and $\check{\mathbf{s}}_q$. The establishment of **Theorem 1** demonstrates the fulfillment of the sliding mode error convergence objectives (O1') and (O2'), and thus the tracking control objectives (O1) and (O2).

6. Simulation study

6.1. Parameter selection

To verify the effectiveness of the dual-loop dual-rate hierarchical control design proposed in this paper, simulation studies are taken on a quadrotor model with the physical parameters listed in **Table 1**.

In terms of the selection of user-defined parameters and sampling rates, the stability conditions (C1) and (C2) obtained in Section 5 play important roles. The followings are steps for parameter selection:

Step 1: Choose the stability regions for tracking errors and estimation errors. Determine the terminal region of the tracking error ρ_s and the desired convergence step size ϵ . Choose λ_p and λ_q for the Hurwitz polynomials. Choose the adaptive estimation gains κ_p and κ_q appropriately so that the bias μ can be small.

Step 2: Determine the minimum sampling period for the outer-loop according to the required computation time for solving the MPC problem and the scaling ratio N between the two sampling periods. Thereafter, pose a lower bound on the sampling period of the inner-loop, denoted as $\Delta_{\min}$.

Step 3: Choose a candidate for the singular perturbation parameter ϵ . Calculate the Lipschitz constants and determine the values of γ_p , γ_q , ζ and η . By applying the graphical method shown in **Fig. 3**, we have that there away exists a maximal sampling period $\Delta_{\max}$ for a given c_p such that (C2) is satisfied. Furthermore: the larger c_p is, the larger the value of $\Delta_{\max}$. Finally, ensuring $\Delta_{\max} > \Delta_{\min}$ delivers an admissible minimum of c_p .

Step 4: As c_p is selected within the admissible range and c_q is selected in the same order of c_p , the maximum of the singular perturbation parameter $\epsilon_{\max}$ can be obtained by the graphic

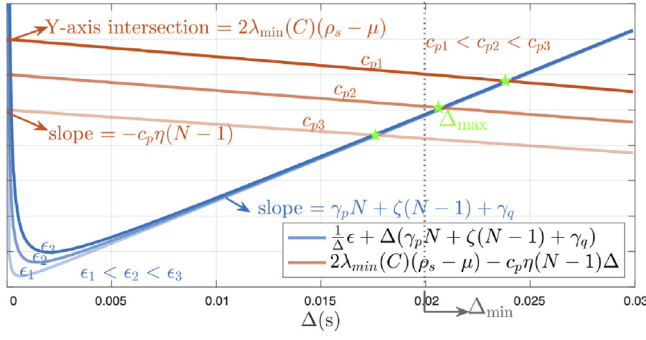


Fig. 3. Graphical expression of stability condition (C2).

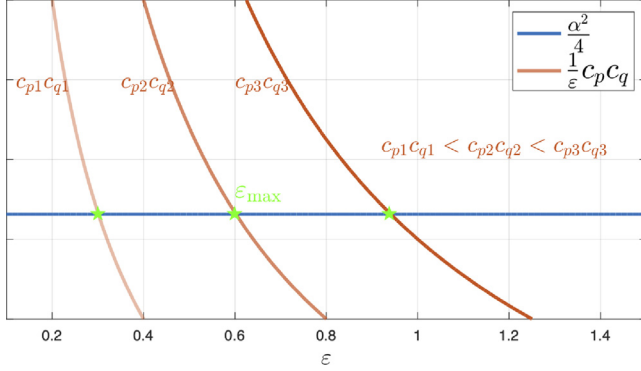


Fig. 4. Graphical expression of stability condition (C1).

expression of (C1) as shown in Fig. 4. Evaluate whether the candidate of ϵ selected in Step 3 is less than $\epsilon_{\max}$. If so, the procedure is ended, otherwise go back to Step 3. and redo the selection.

Following the above steps, we choose $\rho_p = 0.5$, $\rho_q = 0.01$, $\rho_{\sigma_p} = \rho_{\sigma_{p0}} = \rho_{\sigma_{q0}} = 0.01$, $\rho_{\sigma_q} = 0.0001$, $\rho_s = 0.4$, $\epsilon = 0.001$, $\lambda_p = 0.5$, $\lambda_q = 1$, $\kappa_p = 0.1$, $\kappa_{p0} = 1$, $\kappa_q = 0.001$, $\kappa_{q0} = 5$. Then, $\mu \approx 0.06$. A feasible time interval for solving the MPC optimization problem is 0.1 s and N is selected as 5. Thus, the sampling period of the inner-loop is selected as the minimal feasible value – that is $\Delta = 0.02$ s. The singular perturbation parameter is chosen as $\epsilon = 0.2$. By calculating the Lipschitz constants, we obtain that $\alpha \approx 10.2$, $\gamma_p \approx 22.99$, $\zeta \approx 5.2$, $\eta \approx 1.54$, $\gamma_q \approx 19.52$. We then have that the control gain c_p should be at least 7 for ensuring the sufficient stability conditions.

Remark 5. Considering that the stability conditions we obtained in the closed-loop analysis are sufficient but not necessary, we still need to further tune the control parameters according to the actual response in simulations. The conservatism of the theoretical analysis mainly results from the calculation of Lipschitz constants for the nonlinear dynamics. The constants γ_p , ζ , η , and γ_q could be large since they are obtained along the boundaries of the stability regions, leading to an over large slope of the blue line in Fig. 3. Therefore, the actual admissible region of the control gain can be further expanded in practice.

Remark 6. Notice that the convergence regions we choose for calculating the Lipschitz constants are small, which intends to mitigate the conservatism of Lipschitz approximations. The actual regions of convergence could be much larger than the theoretical values, which will be investigated through simulation experiments later in Sections 6.2.1 and 6.2.2.

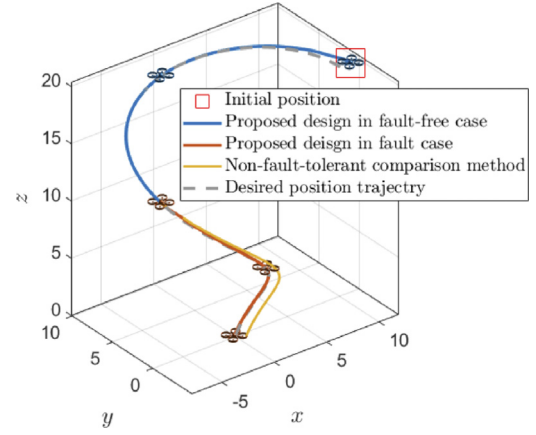


Fig. 5. Translational motion of the quadrotor.

6.2. Simulation results

In the simulations, the desired flight trajectory to track by the quadrotor is set as a downward spiral from the initial position to the base point on the ground ($[0 \ 0 \ 0]^T$). The initial conditions of the quadrotor system are $\mathbf{p}(0) = [11 \ 0 \ 20]^T$ and $\mathbf{q}(0) = [0 \ 0 \ 0]^T$, $\mathbf{v}(0) = [0 \ 2 \ 0]^T$ and $\boldsymbol{\omega}(0) = [0 \ 0 \ 0]^T$. The initial values of the adaptive estimates are selected as $\hat{\sigma}_p(0) = m$, $\hat{\sigma}_{p0}(0) = [0 \ 0 \ 0]^T$, $\hat{\sigma}_q(0) = \mathbf{I}$, $\hat{\sigma}_{q0}(0) = [0 \ 0 \ 0]^T$. The input constraints are set as $\bar{T} = 2 \text{ N}$ and $\bar{\tau} = 2 \text{ N}\cdot\text{m}$.

The control parameters are chosen as: $\lambda_p = 0.5$, $\lambda_q = 1$, $\kappa_p = 0.1$, $\kappa_{p0} = 1$, $\kappa_q = 0.001$, $\kappa_{q0} = 5$, $\epsilon = 0.2$, $N = 5$, $c_p = c_q = 7$. The prediction horizon of the MPC problem is selected as $N_p = 4$. The weighting matrices and scalar in the objective function are $\mathbf{Q} = \text{diag}\{10, 10, 20\}$, $\mathbf{R}_q = \text{diag}\{10, 10, 10\}$ and $R_T = 0.1$.

6.2.1. Fault tolerant capability

To manifest the fault-tolerant capability of the proposed design, we set up the simulation as

- **Fault-free case:** When $0 \leq t < 15$ s, the quadrotor operates in a fault-free case with all the four rotors working properly.
- **Actuator fault case:** When $15 \leq t < 30$ s, an over-voltage failure happens on two of the rotors, causing a sudden increase of 1 N on the rotor's thrust force T_1 and T_4 . Also, a loss-of-effectiveness fault takes place, causing a 10% reduction on forces and torques.
- **Comparison method:** The performance of a control design without the adaptive fault parameter estimator is also tested for comparison.

The simulation results are shown in Figs. 5–8, where system responses in fault-free and faulty cases are indicated by the blue and orange lines, respectively. Figs. 5 and 6 show the translational and rotational motions of the quadrotor. It can be seen that the quadrotor tracks the desired trajectory with high accuracy. At 15 s, when the actuator fault happens, the tracking performance is slightly degraded but then recovers in a short time. The compensation for the unexpected faults is achieved by the adaptive parameter estimation as shown by Fig. 7. The left half of Fig. 8 shows the control commands generated by the controllers and the right half shows the force and torques that are actually applied to the quadrotor. It is obvious that the input constraints are satisfied at all times. The tracking performance of the comparison method is indicated by the yellow line in Fig. 5. Without the adaptive estimator, the control performance obviously deteriorates after the fault occurs at 15 s and gets

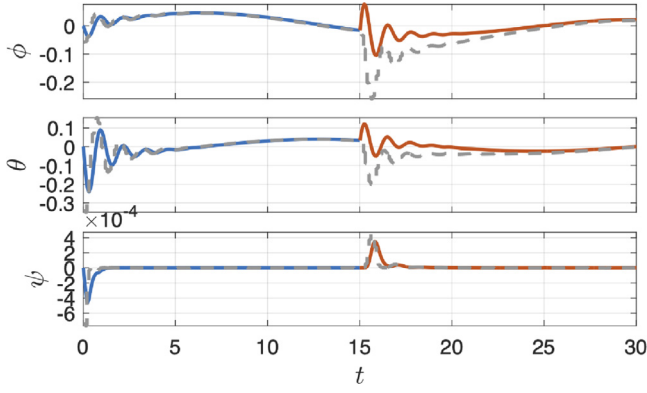


Fig. 6. Rotational motion of the quadrotor.

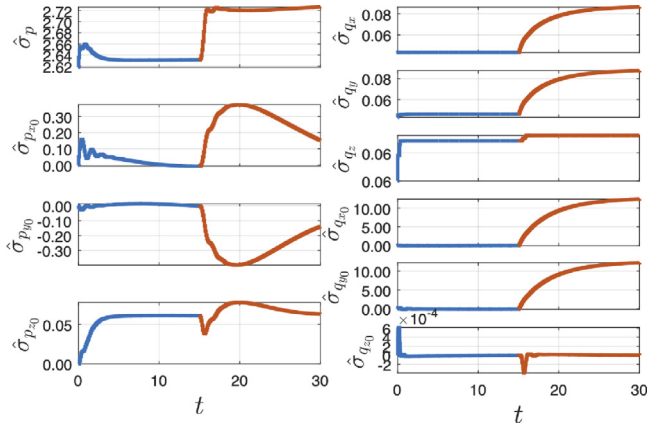


Fig. 7. Online parameter estimation.

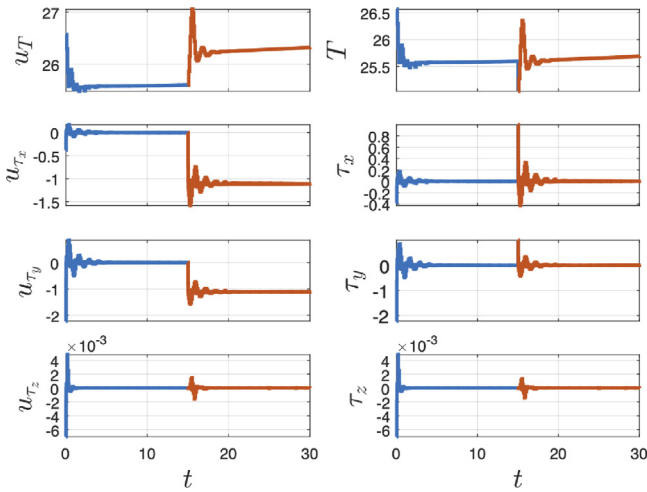


Fig. 8. Control commands (left); Actually applied force and torques (right).

even worse over time. Therefore, by this simulation test, we can verify the effectiveness of the proposed design in terms of fault tolerance.

From the above simulation studies, we verify the fault-tolerant control performance of the proposed design in the presence of loss-of-effectiveness and additive faults. Furthermore, simulations with loss-of-effectiveness and additive faults in different degrees are conducted to investigate the maximum tolerable capability. As a result, the maximal tolerable multiplicative fault

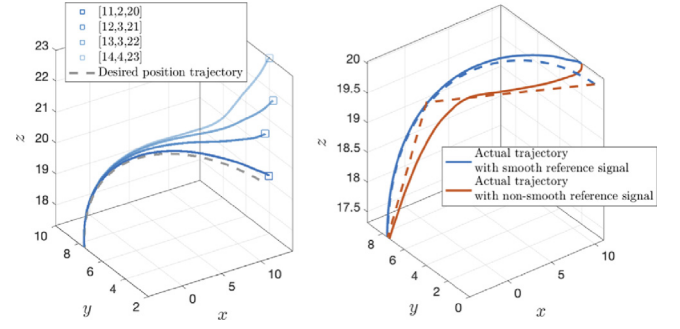


Fig. 9. Tracking performance with different initial positions (left); Tracking performance with smooth and non-smooth reference trajectories (right).

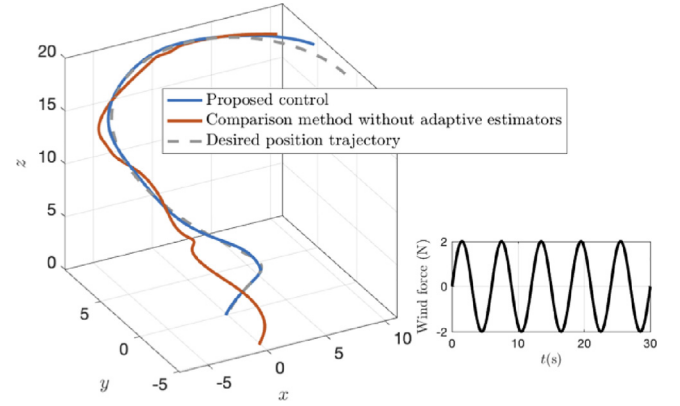


Fig. 10. Tracking performance under wind gust.

is up to 17%. And the developed fault-tolerant scheme can compensate for at most a sudden change in the thrust force of 3.2 N or in the rotation torque of 1.6 N·m.

6.2.2. Region of convergence

In this test, trajectory tracking performances with different initial conditions are also evaluated. The obtained simulation results are given by the left half of Fig. 9. It can be seen that vehicles starting far away from the desired trajectory can still fulfill the tracking control objective, which implies that the actual regions of convergence are larger than those we selected for theoretical analysis. Furthermore, we test the tracking performance for smooth and non-smooth reference signals, shown by the right half of Fig. 9. Both types of trajectories can be tracked under the control of the proposed design.

6.2.3. Adaptability to external disturbances

The adaptability of the proposed design to external disturbances is also studied. A gust of wind is simulated and added to the quadrotor dynamics. The obtained result is shown in Fig. 10, in which the red line represents the tracking performance of a comparison method without the adaptive estimation mechanism. Fig. 10 shows the adaptability of our method for external disturbances.

6.2.4. Adaptability to parameter changes

The proposed adaptive control scheme can also accommodate the changes in the vehicle's mass and the moment of inertia. These model parameters are included in the uncertain parameters σ_p , σ_{p0} , σ_q and σ_{q0} , thereby being estimated by the developed

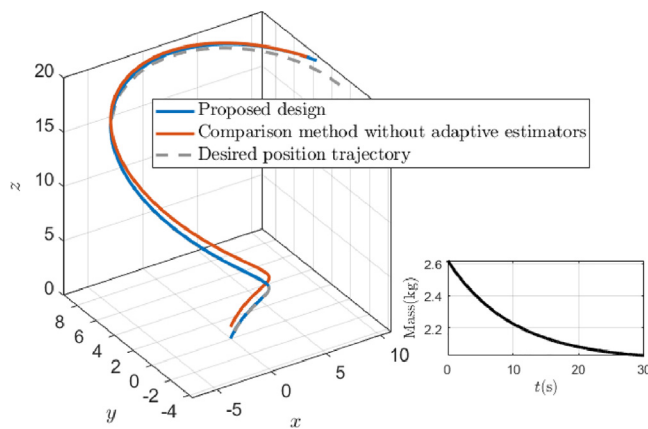


Fig. 11. Tracking performance with mass changing.

adaptive laws. To simulate the practical situation of fuel consumption, tests are conducted when the mass of the vehicle keeps reducing during the flight. Results are shown in Fig. 11, which verifies the adaptability to model uncertainties.

7. Conclusions

This paper presents a dual-loop and dual-rate hierarchical control design for quadrotors subject to unexpected actuator faults. An adaptive parameter estimator and a Lyapunov-based MPC framework are integrated to develop the outer-loop translation controller. Singular perturbation theory is applied to legitimize the dual-loop design and explicitly characterize the high-gain property of the inner-loop rotation controller. The developed translation and rotation controllers are all implemented in a sample-and-hold fashion with different sampling periods. We provide a rigorous stability analysis of the sampled-data and dual-rate control systems with explicitly characterized stability conditions. Comprehensive simulation studies verify the effectiveness of the proposed dual-rate hierarchical fault-tolerant control design.

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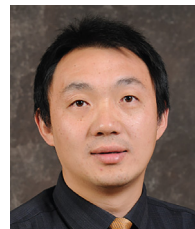
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